

Confidence interval for difference of two means; independent samples, variances unknown but assumed to be equal

Apples example

Background You have the same datasets from the lesson.

Task 1 Calculate the 90% confidence interval

Task 2 Compare the result with the 95% confidence interval from the lesson

Solution:

	NY apples	LA apples
\$	3.80	\$ 3.02
\$	3.76	\$ 3.22
\$	3.87	\$ 3.24
\$	3.99	\$ 3.02
\$	4.02	\$ 3.06
\$	4.25	\$ 3.15
\$	4.13	\$ 3.81
\$	3.98	\$ 3.44
\$	3.99	
\$	3.62	

	NY	LA
Sample mean	\$ 3.94	\$ 3.25
Sample std	\$ 0.18	\$ 0.27
Sample size	10	8
Pooled variance	0.05	
Pooled std	0.22	
90% CI, $t_{16,0.05}$	1.75	

Task 1:

	T	CI low	CI high
90%	\$ 0.51	\$ 0.88	

Task 2: A lower confidence results in a narrower interval (see Margin of error lesson)